CH 3 - Inflation

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Supply constraing → supply chain disruptions

Excess demand → savings that consumer were able to spend with government subsidies to people

3.1 Consumer Price Index

> 📘 **Definition — Consumer Price Index**
>
> The **CPI** measures the **average** change over time in the prices paid by households for a **representative** basket of goods and services, weighted according to aggregate consumer expenditure patterns
>
> The **CPI** is a **pure** price change
>
> The representative basket are 600-700 goods filtered into 8 categories that are obtained from screening census and survery of consumer expenditure.
>
> - Shelter remain the largest sector

Bank of Canada aims 2% CPI inflation annually. Adjustments of tax brackets and wages occured based on CPI change.

Data are collected from scanner data, manual collection, administration data, and web scrapping

Housing in CPI in Canada:

Owned accomodation: mortgage interest, ropeerty tax, insurance, depreciation, etc.

Rent directly included

Category's contribution to CPI inflation:

$$\text{basket weight} \times \text{category inflation}$$

3.2 Inflation

> 📘 **Definition — Inflation**
>
> **Inflation** is a sustained increase in the average price level
> $\pi =$ inflation
> $\text{CPI}_t =$ level of the CPI at time t
> $\pi_t =$ year over year change in the CPI
> $t =$ months
> $\text{CPI}_{t-12} =$ Level of the CPI at time $t-12$
>
> $$\pi_t = \frac{\text{CPI}_t - \text{CPI}_{t-12}}{\text{CPI}_{t-12}} \times 100$$

Example:

Year	Price	Inflation
0	10000	
1	15000	50%
2	15000	0%
...	...	0% onwards

Notice that even though inflation becomes 0 after year 2, but the price remains constant after inflation. Thus inflation rate does not always necessarily mean a good thing as the price becomes higher overall.

> 📘 **Definition — Deflation, Disinflation**
>
> **Deflation** is a sustained decrease in the average price level
>
> **Disinflation** is a decline in the inflation rate while it remains positive. The price level contunues to rise but slowly. That is inflation rate is decreasing, or the price is increasing in a decreasing rate

Example:

Good	Price in 2024	Price in 2025	Basket quantity
G	5	5.2	2
W	4	4.1	3
C	4	4.06	4

Basket cost of CPI in 2024 $= P_G \times Q_G + P_W \times Q_W + P_C + Q_C = 38$

Basket cost of CPI in 2025 $= P_G \times Q_G + P_W \times Q_W + P_C + Q_C = 38.94$

$\text{CPI}_{2024} = 100$, as a reference year

$\text{CPI}_{2025} = \left(\frac{38.95}{38}\right) \times 100 = 102.5$, based on 2024 as a reference year

$\pi_{2025} = \frac{102.5-100}{100} \times 100 = \frac{38.94-38}{38} = 2.5\%$, inflation rate is 2.5%

Base Year effect: occurs when an unusual price movement wewelves months earlier affects the current year-over-year inflation rate.

- An unusually low CPI 12 months ago raises the current year-to-year rate
- An unusually high CPI 12 months ago lowers the current year-to-year rate